

Nyasha Hama

Software Engineer | Product and Backend Systems

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Summary

Software engineer building operational products from interface to data model, with particular attention to retries, access boundaries, and recoverable failures. Work spans ClinicPulse, StrataHQ, Rust failure-testing tooling, and maintainer-merged changes to Turso and CrossHair.

Current confidential full-stack work includes C#; client and implementation details are withheld under NDA.

Experience

Self-Directed Software Engineer, Independent engineering – Cape Town, South Africa Jan 2023 – present

- Designed and built independent operational software across Next.js interfaces, Go services, PostgreSQL data models, and automated test and delivery pipelines; selected implementations follow.

Selected Projects

ClinicPulse

Alpha clinic-operations demo with field, district, public, and admin workflows.

- Built persisted offline field reporting across a Next.js client and Go API, distinguishing safe retries from conflicting submissions and recording sync attempts for review.
- Added regressions for duplicate payloads, uniqueness races, and timestamp precision; built role-aware views backed by PostgreSQL.
- [Source](#) | [Demo](#)

StrataHQ

Seeded beta demo for managing agents, trustees, and residents.

- Implemented Go/PostgreSQL bank-statement imports with row fingerprints, transactional payment application, and manual review of ambiguous matches and overpayments.
- Built scheme-scoped product workflows across levies, maintenance, and governance in Next.js, backed by Go APIs and typed SQL access.
- [Source](#) | [Demo](#)

TxProof

Rust harness for bounded counterexample search against a synthetic payment reference application.

- Generated state-valid failure campaigns for ambiguous provider responses, retries, webhooks, and restart boundaries, with deterministic replay and PostgreSQL invariant checks.
- [Source](#)

Open Source Contributions

Upstream Contributor, Turso May 2026

- Authored two maintainer-merged Rust fixes for schema-transition and sequence-state correctness in a SQLite-compatible engine: [PR #6993](#) | [PR #7117](#)

Upstream Contributor, CrossHair May 2026

- Moved call-target normalization into the C tracer while preserving callable dispatch and error behavior, with regression coverage: [PR #413](#)

Skills

Languages: TypeScript, Go, Rust, SQL, Java, C#, C++

Product and Data: React, Next.js, REST APIs, PostgreSQL, pgx, sqlc, Redis, Spring Boot

Delivery and Verification: Docker, GitHub Actions, Playwright, Vitest, structured logging, Prometheus

Education

University of Johannesburg — Coursework toward BSc Computer Science, completed through final year, 2022–2024